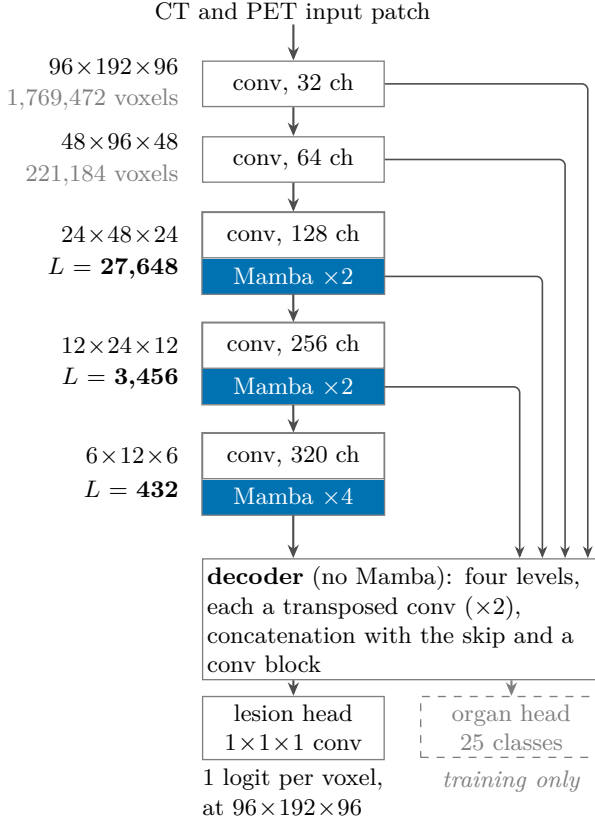


One Mamba block

$$h' = h + \sum_{d=1}^{6} U_d(M_d(F_d(\mathrm{LN}\, h)))$$

h, h': the stage's map before and after the block (same shape).

LN: layer norm over the channels.

F_d: flatten the map to $L = D \cdot H \cdot W$ tokens in scan direction d: $\pm x$, $\pm y$ or $\pm z$ (minus: reversed).

M_d: a Mamba layer with its own weights.

U_d: unflatten to the grid.

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conv: two $3\times3\times3$ convolutions, instance norm, leaky ReLU, residual; stride 2 in every stage but the first.

Mamba $\times n$: n Mamba blocks in a row.

L: sequence length per scan direction. The two shallowest stages are too long to scan.

Without the Mamba blocks this is run 6's CNN-only twin.